

SILISTRA, Bulgaria

WMO: 155500

Lat: 44.117N

Lon: 27.267E

Elev: 52

StdP: 14.67

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.5	17.0	7.5	8.1	13.0	12.5	10.4	18.1	35.9	36.8	31.3	26.2	5.8	50	0.427

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	92.6	74.0	89.6	73.5	86.9	72.1	76.2	88.3	74.7	87.0	73.2	84.9	4.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.4	120.3	83.9	70.6	113.0	81.9	69.2	107.4	80.2	39.7	88.5	38.2	87.4	36.8	85.0	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.4	21.9	18.4	DB	8.3	98.0	5.9	4.5	4.0	101.2	0.5	103.8	-2.8	106.3	-7.2	109.6	
			WB	7.5	78.9	5.7	2.0	3.4	80.4	0.1	81.6	-3.1	82.7	-7.2	84.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.4	32.6	36.9	45.6	55.5	65.1	72.0	76.3	75.9	66.6	55.8	45.9	35.7
	DBStd	16.93	9.21	10.01	8.60	6.37	6.18	5.74	5.01	4.93	5.81	7.33	8.19	8.78
	HDD50	1782	542	375	189	22	1	0	0	0	0	30	174	449
	HDD65	4600	1005	788	601	290	77	10	0	1	51	294	575	908
	CDD50	3761	2	8	53	188	468	660	815	802	497	212	50	6
	CDD65	1103	0	0	0	6	79	220	351	338	98	10	1	0
	CDH74	9038	0	0	3	49	566	1850	3162	2871	502	34	1	0
	CDH80	3235	0	0	0	6	115	638	1284	1094	97	1	0	0
Wind	WSAvg	5.6	5.8	6.6	7.5	6.1	5.7	5.0	4.8	4.2	4.9	4.9	5.3	6.0
Precipitation	PrecAvg	21.2	1.6	1.0	1.5	1.7	2.0	2.4	1.9	1.7	2.2	1.7	1.7	1.8
	PrecMax	36.7	4.6	3.0	3.7	4.1	6.2	5.4	6.1	4.3	6.4	4.3	3.6	5.1
	PrecMin	12.1	0.2	0.0	0.2	0.1	0.5	0.5	0.1	0.2	0.1	0.1	0.0	0.2
	PrecStd	5.5	1.0	0.7	0.8	0.9	1.3	1.2	1.5	1.1	1.7	1.0	1.0	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.6	64.0	72.1	80.0	87.4	92.6	98.7	96.2	87.1	78.1	70.3	61.2
		MCWB	50.8	52.8	56.4	63.5	68.7	74.7	74.0	74.3	69.6	66.6	60.6	53.9
	2%	DB	52.3	58.5	67.4	73.5	83.0	89.3	93.8	91.9	82.8	73.9	65.7	55.7
		MCWB	47.3	50.2	54.5	59.9	68.0	73.3	73.9	74.2	68.2	64.3	58.1	51.0
	5%	DB	48.5	54.0	63.0	69.7	80.1	86.6	90.2	88.9	79.4	69.9	61.1	51.0
		MCWB	44.5	47.3	52.2	58.1	66.9	72.0	73.6	73.5	67.0	62.3	56.0	47.5
	10%	DB	45.0	50.1	58.9	66.3	76.8	83.7	86.9	86.1	76.5	66.8	57.2	47.4
		MCWB	41.9	44.8	50.0	56.0	65.0	70.4	72.4	71.9	65.8	60.5	53.3	44.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.6	53.4	57.7	64.3	72.4	77.0	77.7	79.1	71.3	68.1	61.6	54.5
		MCDB	57.0	62.3	70.3	76.9	82.7	88.9	90.5	89.1	83.1	76.1	69.1	59.8
	2%	WB	47.6	50.4	54.9	61.1	69.6	74.8	76.0	76.5	69.4	65.0	58.8	51.0
		MCDB	51.5	57.4	65.5	71.6	80.6	86.9	88.8	87.9	80.2	71.7	64.7	55.4
	5%	WB	44.9	47.8	52.8	58.9	67.6	72.9	74.7	74.5	67.8	62.9	56.3	48.1
		MCDB	48.3	53.7	62.4	68.0	78.0	84.9	87.4	86.4	77.7	69.0	60.7	51.0
	10%	WB	41.9	45.0	50.4	57.0	65.6	71.2	73.2	72.9	66.4	60.6	53.6	44.8
		MCDB	44.9	49.7	58.0	64.7	75.6	82.4	85.1	84.2	75.2	66.4	57.2	47.3
Mean Daily Temperature Range	5% DB	MDBR	9.0	11.5	14.6	16.3	17.6	17.3	17.1	17.1	14.9	12.9	11.0	9.0
		MCDBR	14.1	18.2	21.7	21.0	22.0	21.4	22.2	20.8	19.2	17.3	16.9	14.8
		MCWBR	10.4	12.0	12.0	11.2	11.3	9.9	9.2	9.1	8.9	10.3	11.0	11.3
	5% WB	MCDBR	13.4	18.0	20.6	19.0	20.0	20.1	19.7	19.3	17.7	16.3	16.0	14.4
		MCWBR	10.4	12.2	11.9	10.7	10.8	10.2	9.2	9.2	9.0	10.4	10.6	11.4
Clear-Sky Solar Irradiance	taub	0.348	0.379	0.414	0.464	0.438	0.444	0.456	0.469	0.437	0.398	0.371	0.350	
	taud	2.285	2.212	2.166	2.035	2.191	2.211	2.189	2.144	2.198	2.292	2.314	2.303	
	Ebn at Noon	238	251	258	254	266	264	259	250	247	242	227	224	
	Edh at Noon	27	35	42	51	45	44	45	45	40	32	27	24	
All-Sky Solar Radiation	RadAvg	443	693	1065	1496	1881	2044	2068	1852	1345	858	488	374	
	RadStd	63	100	107	158	130	141	128	100	102	93	62	48	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (48 neighbors)	+0.85	N/A	N/A	N/A	N/A	N/A	N/A	-214	+157	+99

Nomenclature: See separate page